

SEDALIA, MO, USA

WMO: 724453

Lat: 38.704N

Lon: 93.183W

Elev: 274

StdP: 98.07

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 03994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-12.0	-19.7	0.7	-13.6	-17.3	0.8	-10.9	12.1	4.1	11.1	3.3	3.4	320	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	35.8	24.3	33.8	24.3	32.4	24.2	26.4	32.4	25.7	31.5	25.0	30.7	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.5	30.1	23.9	19.5	29.3	23.1	18.4	28.2	83.9	32.2	80.6	31.7	77.4	30.7	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	8.3	-18.4	37.3	2.7	2.3	-20.3	38.9	-21.9	40.2	-23.4	41.5	-25.3	43.1		
(5)			-18.9	27.7	2.6	0.6	-20.7	28.2	-22.2	28.5	-23.6	28.9	-25.5	29.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	0.0	2.1	7.9	13.6	18.9	23.7	26.0	25.2	21.1	14.4	8.0	1.9
	DBStd	10.45	6.52	6.61	6.25	5.22	4.43	3.38	3.17	3.20	3.99	5.25	5.77	6.12
	HDD10.0	1067	316	232	114	24	1	0	0	0	0	19	102	259
	HDD18.3	2543	569	454	326	157	49	3	0	1	18	144	311	511
	CDD10.0	2389	4	11	50	132	277	412	496	471	332	154	43	7
	CDD18.3	824	0	0	3	15	67	166	237	213	99	22	2	0
	CDH23.3	7979	0	1	22	122	533	1568	2562	2131	869	162	8	0
	CDH26.7	3266	0	0	2	19	144	608	1179	961	316	37	0	0
Wind	WSAvg	4.0	4.6	4.6	4.7	4.7	4.0	3.8	3.1	3.1	3.3	3.9	4.4	4.4
Precipitation	PrecAvg	1072	43	49	72	118	142	124	113	85	109	86	78	54
	PrecMax	1527	140	130	164	323	389	327	341	242	337	255	258	142
	PrecMin	677	8	0	25	30	29	33	12	32	23	11	22	14
	PrecStd	212	28	33	35	49	78	69	85	46	73	54	59	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	22.4	26.0	28.8	31.8	35.2	38.6	37.9	34.2	30.0	24.5	19.8
		MCWB	12.1	14.6	17.4	19.5	22.6	24.9	24.3	23.7	23.3	21.7	16.7	15.4
	2%	DB	15.6	18.3	23.0	26.8	29.9	33.2	36.1	35.8	32.2	27.5	22.0	17.0
		MCWB	10.8	12.4	15.6	18.0	21.7	24.3	24.7	23.9	23.2	19.7	15.3	13.2
	5%	DB	12.4	15.6	20.8	24.4	28.2	32.0	33.9	33.6	30.2	24.9	19.1	13.4
		MCWB	7.8	10.9	14.4	16.6	21.1	24.1	24.8	24.0	22.3	18.1	14.0	9.5
	10%	DB	9.1	12.2	17.8	22.5	26.6	30.2	32.4	31.8	28.1	22.7	16.9	11.0
		MCWB	5.6	8.1	12.6	15.8	20.0	23.2	24.9	23.9	21.2	17.6	12.4	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	16.1	18.7	20.7	24.3	26.6	27.5	27.1	25.3	23.0	18.4	16.6
		MCDB	17.6	20.3	24.5	27.2	29.9	32.4	33.2	32.9	31.4	27.9	22.4	19.7
	2%	WB	11.4	13.8	17.0	19.3	23.1	25.6	26.7	26.1	24.1	21.4	16.5	13.7
		MCDB	14.3	17.3	21.6	24.7	28.4	31.3	32.6	32.0	29.8	25.2	20.0	16.4
	5%	WB	8.2	11.1	15.4	18.0	21.9	24.8	25.9	25.2	23.4	19.8	14.8	10.5
		MCDB	11.9	15.1	19.6	22.9	26.8	30.2	31.9	31.1	28.5	23.2	18.3	12.8
	10%	WB	5.8	7.9	13.1	16.7	20.9	23.9	25.3	24.4	22.5	18.3	13.0	7.8
		MCDB	9.1	11.9	17.2	21.3	25.1	29.1	31.1	30.0	26.9	21.9	16.3	10.2
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.7	10.5	10.9	10.4	10.1	10.3	10.9	11.4	10.8	10.0	8.7
		MCDBR	13.7	14.2	13.5	13.4	11.9	11.4	11.8	13.0	12.7	12.7	12.8	12.2
		MCWBR	9.6	9.7	7.8	7.5	6.1	5.1	4.2	4.6	5.5	6.5	8.2	8.9
	5% WB	MCDBR	12.6	13.5	11.6	11.3	10.7	10.2	10.3	10.9	11.0	10.6	11.3	11.4
MCWBR		9.6	10.1	7.6	7.3	6.1	5.3	4.8	4.9	5.3	6.6	8.5	9.3	
Clear-Sky Solar Irradiance	taub	0.297	0.311	0.337	0.388	0.424	0.427	0.446	0.450	0.387	0.345	0.307	0.293	
	taud	2.466	2.427	2.410	2.279	2.241	2.266	2.226	2.223	2.361	2.466	2.517	2.502	
	Ebn at Noon	884	917	923	889	859	854	834	821	857	866	867	865	
	Edh at Noon	80	95	107	130	138	135	139	136	112	91	76	72	
All-Sky Solar Radiation	RadAvg	2.16	2.88	3.80	4.92	5.68	6.42	6.39	5.62	4.82	3.42	2.35	1.86	
	RadStd	0.25	0.33	0.34	0.32	0.50	0.53	0.41	0.32	0.31	0.36	0.25	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page